

REFLECTIVE OPTICAL PHOTOPLETHYSMOGRAPHY (cPPG)

Mathematical Formulas, Signal Processing Pipeline & Peer-Reviewed Scientific Validation

Project: SMRUTI Vitality Health Checkup Suite | **Sensing Modality:** Camera-based Reflective PPG (cPPG) | **Release:** Technical & Scientific Analysis Report **Scope:** Mathematical inventory of all algorithmic formulas, source files, physiological foundations, and direct literature citations.

1. System Architecture & The Optical Extraction Process

Hardware Isolation & The Sensor Transition: Modern smartphone fingerprint sensors operate exclusively within a hardware-isolated Trusted Execution Environment (TEE). Third-party user-space applications are cryptographically prohibited from accessing raw capacitance matrices or optical sensor frames. To deliver non-invasive health screening on commodity smartphones, this system transitions to Reflective Optical Photoplethysmography (cPPG) using the rear CMOS camera sensor combined with the adjacent continuous LED flash.

Stage	Module / File	Engineering Operation	Physiological / Signal Objective
1. Optical Capture	PulseSheet.tsx CameraView	Captures camera frames at 10-15 FPS with flash enabled, skipProcessing=true.	Transilluminates capillary beds of fingertip dermis; captures time-series optical backscatter.
2. Tissue Verification	pulse.ts frameBrightness()	Subsamples frame red and green channels. Enforces $avgRed \geq 95$ and $avgRed \geq 1.15 * avgGreen$.	Prevents mathematical aliasing from ambient AC lighting.
3. Temporal Resampling	pulse.ts resample()	Linear interpolation of timestamped luminance samples onto a uniform 20 Hz temporal grid.	Eliminates capture jitter, fulfilling the strict equidistant time requirement for digital filtering.
4. Outlier Clamping	pulse.ts clampMotionSpikes()	Derivative thresholding: clamps step changes exceeding 3.5x the median absolute first difference.	Suppresses mechanical motion artifacts caused by transient finger slip.

Stage	Module / File	Engineering Operation	Physiological / Signal Objective
5. Bandpass Filter	pulse.ts applyBandpass()	Zero-mean centering + 2nd-order Butterworth cascaded highpass (0.7 Hz) and lowpass (3.5 Hz).	Isolates cardiac pulsatile frequencies; rejects DC respiratory baseline wander and noise.
6. Optical Inversion	pulse.ts analyzeSignalWindow()	Signal polarity inversion: $s[n] = -\text{filtered}[n]$.	Beer-Lambert Law alignment: systolic capillary blood expansion absorbs more light; inversion makes pulses positive peaks.
7. Peak & Beat Detection	pulse.ts analyzeSignalWindow()	Local maxima search with refractory distance constraint ($\geq 0.35\text{s}$) and adaptive prominence.	Extracts true systolic peaks while preventing double-counting of dicrotic notches.
8. Interval Filtering	pulse.ts analyzeSignalWindow()	Derives RR intervals; rejects intervals with instantaneous BPM outside $[40, 200]$.	Guarantees high-fidelity Inter-Beat Interval (IBI) series for downstream autonomic HRV computation.

2. Mathematical Formulas & Algorithmic Formulations

Every formula used across the mobile application and the research testbench is itemised below with exact parameter definitions. Sampling rate is $f_s = 20\text{ Hz}$ after resampling; N denotes a sample count and M the number of usable RR intervals.

2.1 Optical Tissue Verification & Subsampled Luminance

The decoded frame is an RGBA buffer, so channel c of pixel k lies at byte $4k + c$. With a subsample factor s :

$$\bar{I}_{\text{red}} = \frac{1}{N} \sum_{k=0}^{N-1} I[4sk], \quad \bar{I}_{\text{green}} = \frac{1}{N} \sum_{k=0}^{N-1} I[4sk + 1]$$

A frame is accepted as fingertip tissue only when both conditions hold:

$$\text{valid} \iff (\bar{I}_{\text{red}} \geq 95) \wedge (\bar{I}_{\text{red}} \geq 1.15 \bar{I}_{\text{green}})$$

2.2 Uniform Temporal Resampling (Linear Interpolation)

For a target instant t_k bracketed by acquired samples $t_i \leq t_k < t_{i+1}$:

$$y(t_k) = y(t_i) + [y(t_{i+1}) - y(t_i)] \cdot \frac{t_k - t_i}{t_{i+1} - t_i}$$

2.3 Derivative-Based Outlier Clamping (Motion Artefact Reduction)

$$\Delta x[i] = |x[i] - x[i-1]|, \quad T = \max(3.0, 3.5 \cdot \text{median}_i(\Delta x[i]))$$

$$x^*[i] = \begin{cases} x^*[i-1] + \text{sgn}(x[i] - x^*[i-1]) \cdot T, & |x[i] - x^*[i-1]| > T \\ [6pt] x[i], & \text{otherwise} \end{cases}$$

2.4 Second-Order Butterworth Bandpass IIR Filter (Biquad Formulation)

With cutoff f_c , damping $d = \sqrt{2}$, and passband 0.7–3.5 Hz:

$$\theta = \frac{\pi f_c}{f_s}, \quad \beta = \frac{1}{2} \cdot \frac{1 - \frac{d}{2} \sin \theta}{1 + \frac{d}{2} \sin \theta}, \quad \gamma = \left(\frac{1}{2} + \beta\right) \cos \theta$$

The cascaded sections are applied through the standard difference equation:

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2]$$

2.5 Heart Rate (BPM) from Peak Intervals

For detected peak indices $p[k]$:

$$RR[k] = \frac{p[k] - p[k-1]}{f_s} \text{ [s]}, \quad \text{HR} = \text{round}\left(\frac{60}{\text{median}(RR_{\text{clean}})}\right) \text{ [bpm]}$$

2.6 Heart Rate Variability: RMSSD, SDNN, pNN50 & pNN20

$$\text{RMSSD} = 1000 \sqrt{\frac{1}{M-1} \sum_{k=1}^{M-1} (RR[k+1] - RR[k])^2} \text{ [ms]}$$

$$\text{SDNN} = 1000 \sqrt{\frac{1}{M} \sum_{k=1}^M (RR[k] - \overline{RR})^2} \text{ [ms]}$$

$$\text{pNN}_x = \frac{|\{k : |RR[k+1] - RR[k]| > x \text{ ms}\}|}{M-1} \times 100 \text{ [\%]}, \quad x \in \{20, 50\}$$

2.7 Autonomic Stress Score

$$S = \text{clamp}\left(15 + 1.8 \cdot \max(0, 50 - \text{RMSSD}), 10, 95\right) \text{ [\%]}$$

2.8 Systolic Upstroke Time (Foot-to-Peak) & Vascular Elasticity

$$j_{\text{foot}} = \arg \min_{j \in [p - \lfloor 0.4 f_s \rfloor, p]} s[j], \quad T_{\text{up}} = \frac{p - j_{\text{foot}}}{f_s} \times 1000 \text{ [ms]}$$

Optimal elasticity: $T_{\text{up}} \geq 130 \text{ ms}$ Elevated stiffness: $T_{\text{up}} < 105 \text{ ms}$

2.9 Blood Oxygen Saturation (SpO₂ via Ratio of Ratios)

$$R = \frac{AC_{\text{red}}/DC_{\text{red}}}{AC_{\text{green}}/DC_{\text{green}}}, \quad \text{SpO}_2 = \text{round}(110 - 25R) \text{ [\%]}$$

2.10 LF/HF Ratio (Frequency-Domain HRV Surrogate)

$$\sigma_{LF}^2 = \max(0, \text{SDNN}^2 - \text{RMSSD}^2), \quad \sigma_{HF}^2 = \text{RMSSD}^2$$

$$\frac{LF}{HF} = \frac{\sigma_{LF}^2}{\sigma_{HF}^2}$$

2.11 Stroke Volume, Cardiac Output & Cardiac Workload

$$PP = BP_{\text{sys}} - BP_{\text{dia}} \text{ [mmHg]}, \quad SV = \text{round}(1.5 \cdot PP) \text{ [mL]}$$

$$CO = \frac{SV \cdot \text{HR}}{1000} \text{ [L/min]}, \quad RPP = \text{HR} \cdot BP_{\text{sys}} \text{ [mmHg} \cdot \text{bpm]}$$

3. Scientific Validation & Direct Research Paper Citations

The mathematical principles, optical filter thresholds, and physiological interpretations implemented in the codebase are grounded in and validated by peer-reviewed biomedical engineering literature:

Biomarker / Formula	Physiological Grounding	Primary Literature Citation
Photoplethysmography & Beer-Lambert Law	Volumetric expansion of arteriolar vessels during ventricular systole increases light absorption by oxy- and deoxyhemoglobin.	Allen, J. (2007). <i>Physiological Measurement</i> , 28(3), R1-R39.
Smartphone Camera PPG (cPPG)	CMOS image sensors capture micro-fluctuations in reflected red/green luminance caused by pulsatile dermal perfusion.	Scully, C. G., et al. (2012). <i>IEEE TBME</i> , 59(2), 303-306.
Butterworth Bandpass (0.7-3.5 Hz)	Isolates cardiac fundamental and harmonic frequencies (42-210 BPM) while stripping respiratory drift (<0.5 Hz) and sensor noise.	Elgendi, M. (2012). <i>Current Cardiology Reviews</i> , 8(1), 14-25.
Time-Domain HRV (RMSSD, SDNN, pNN50, pNN20)	RMSSD and pNN20/50 index high-frequency parasympathetic (vagal) cardiac control. SDNN indexes total autonomic regulatory capacity.	Task Force of ESC & NASPE (1996). <i>Circulation</i> , 93(5), 1043-1065. Shaffer, F. et al. (2017). <i>Frontiers in Public Health</i> , 5, 258.

Biomarker / Formula	Physiological Grounding	Primary Literature Citation
LF/HF Ratio	Quantifies the sympathovagal balance, acting as a clinical surrogate for physical and mental stress states.	Malliani, A., et al. (1991). <i>Circulation</i> , 84(2), 482-492.
Blood Oxygen Saturation (SpO2)	The ratio of pulsatile to baseline light absorption (AC/DC) across Red and Green wavelengths isolates oxygenated hemoglobin saturation.	Karlen, W., et al. (2010). <i>Conference of the IEEE Engineering in Medicine and Biology</i> .
Systolic Upstroke Time (T _{up}) & Elasticity	Moens-Korteweg Law. Vessel stiffening increases pulse wave velocity (PWV), accelerating the systolic upstroke (T _{up} < 105 ms).	Chowienczyk, P. J., et al. (1999). <i>JACC</i> , 34(7), 2007-2014.
Stroke Volume (SV) & Cardiac Output (CO)	Pulse contour analysis models (e.g., Liljestrand-Zander) establish that arterial pulse pressure is proportionally related to stroke volume.	Wang, L., et al. (2018). <i>Sensors</i> , 18(1), 173.
Respiration Rate via RSA	Respiratory Sinus Arrhythmia: Intrathoracic pressure changes during respiration modulate venous return and autonomic vascular tone.	Karlen, W., et al. (2013). <i>IEEE TBME</i> , 60(7), 1946-1953.
Cuff-Anchored Calibrated BP	Optical cPPG cannot derive absolute hydrostatic pressure without a reference anchor. Perturbation models track changes relative to known readings.	Mukkamala, R., et al. (2015). <i>IEEE TBME</i> , 62(8), 1879-1901.

4. Regulatory Boundaries & Software as a Medical Device (SaMD) Notice

Clinical Boundary & Non-Diagnostic Scope: The optical photoplethysmography pipeline documented herein operates exclusively as a **personal wellness and non-invasive physiological screening tool**. It does not measure blood glucose or invasive arterial hemodynamics directly. In accordance with FDA / CE SaMD guidance:

- 1. Metabolic Surrogates:** Advanced Glycation End-products (AGEs) alter arterial wall stiffness and autonomic vagal tone over time, but PPG cannot replace chemical glucose testing. Patients must **never** adjust insulin or hypoglycemic medication based on optical estimations.
- 2. Blood Pressure Grounding:** Blood pressure estimates are one-point cuff-anchored perturbation models, not direct hydrostatic measurements, and require

regular recalibration.

3. **Diagnostic Authority:** Users experiencing symptoms or seeking clinical diagnoses must consult a licensed physician and undergo standardized laboratory blood tests.